



Beacon Free

Kubernetes Version Monitoring for Headlamp

FREE EDITION — USER MANUAL

Infrastructure

Updated: 26/05/2026, 06:41:36

Send Report

Application ↑	Namespace ▾	Current Version	Latest Version	Status ▾
Envoy Gateway	envoy-gateway-system	v1.8.0	v1.9.0-rc.0	Update Available ↗
Kube State Metrics	ops-prometheus	v2.18.0	v2.19.0	Update Available ↗
Prometheus Operator	ops-prometheus	v0.90.1	v0.91.0	Update Available ↗
Reloader	reloader	v1.1.0	v1.4.17	Update Available ↗
Sealed Secrets	sealed-secrets	0.32.2	v0.37.0	Update Available ↗

Version 2.1.0 · May 2026

Headlamp Plugin — Kubernetes cluster monitoring

CONTENTS

- 1 What is Beacon?
- 2 Getting Around
 - The Beacon sidebar
 - Page layout
- 3 Infrastructure Page
 - Reading the version table
 - Sorting and filtering
 - Status badges explained
- 4 Headlamp Plugins Page
 - What is monitored
 - Reading the results
- 5 Applications Page (Pro Feature)
- 6 Settings
 - Infrastructure tab
 - Headlamp Plugins tab
 - Schedule tab
- 7 Frequently Asked Questions

1 What is Beacon?

Beacon is a plugin for Headlamp — the open-source Kubernetes web UI. Beacon adds a dedicated section to Headlamp that tracks the versions of the software running in your cluster and compares them against the latest releases published on GitHub and container registries.

At a glance, Beacon tells you which components are up to date and which ones have updates waiting — so your team always knows the state of the cluster without manually checking each project's release page.

What Beacon Free monitors

SECTION	WHAT IT TRACKS	AVAILABLE IN
Infrastructure	Core Kubernetes tooling: Envoy Gateway, Prometheus, Reloader, Sealed Secrets, etc.	Free
Headlamp Plugins	Headlamp itself and all installed plugins (Flux, Kubescape, Trivy, etc.)	Free
Applications	Your own cluster workloads and custom deployments	Pro only

How Beacon works

Beacon runs a lightweight background job (*beacon-updater*) on a daily schedule. This job reads the list of components you have selected to monitor, looks up their latest published versions, and stores the results in the cluster as a ConfigMap. The Headlamp plugin reads those results and displays them in the version tables you see on screen.

No data leaves your cluster. Beacon only makes outbound requests to public GitHub and container registry APIs to fetch version information.

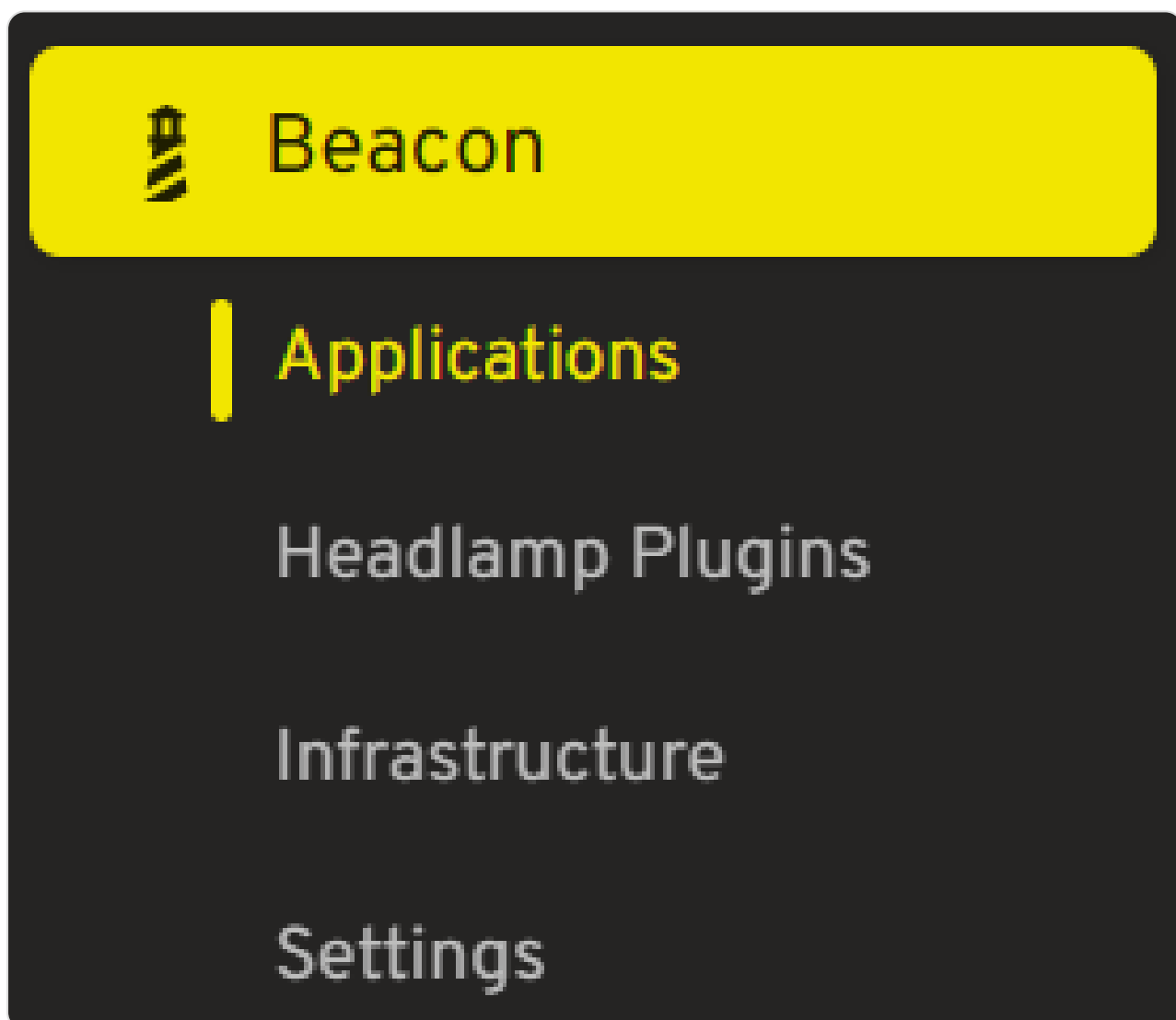
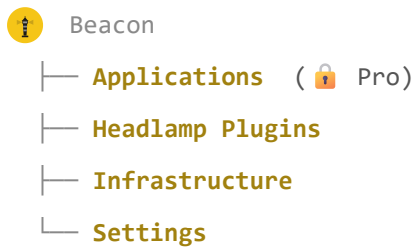


Figure 1 — Beacon appears in the Headlamp left sidebar after installation

2 Getting Around

The Beacon sidebar

After Beacon is installed, a new **Beacon** entry appears in the Headlamp left sidebar. Expand it to reveal four pages:



Clicking **Beacon** in the sidebar lands you directly on the **Infrastructure** page.

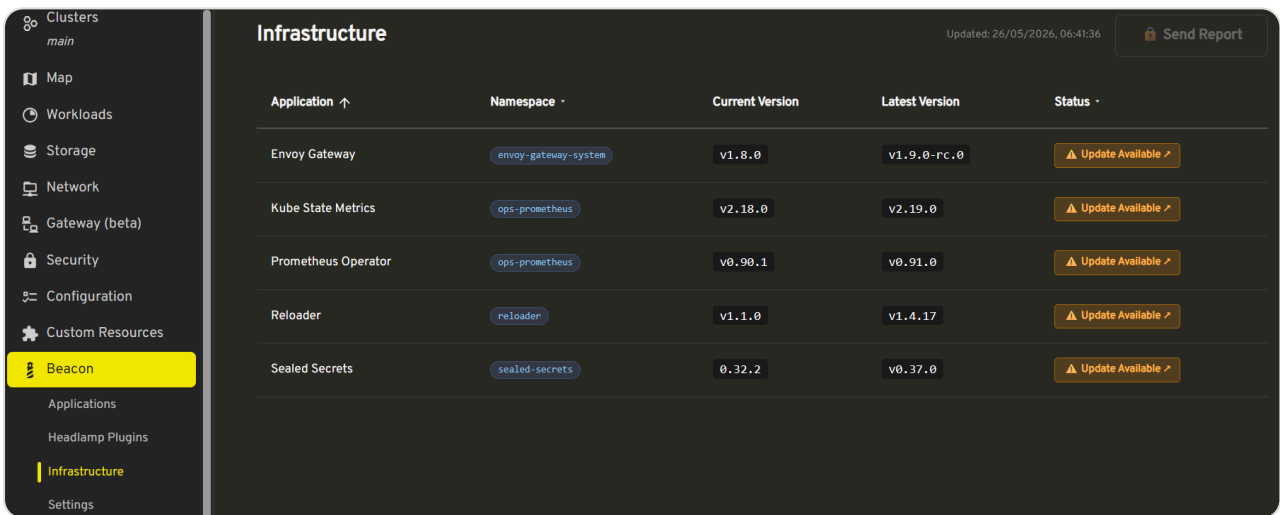


Figure 2 — The four Beacon navigation entries in alphabetical order

Page layout

Every Beacon monitoring page shares the same layout:

AREA	DESCRIPTION
Title (top-left)	Name of the current section (Infrastructure, Headlamp Plugins, or Applications)
Last updated (top-right)	Timestamp of the most recent version fetch
Send Report button	Locked in Free edition (requires Pro)
Version table	One row per monitored component with current and latest versions



First time here?

If you see a "not configured yet" card instead of a version table, go to **Settings**

and use the Infrastructure or Headlamp Plugins tab to scan, select, and save what to monitor. See Section 6 for instructions.

3 Infrastructure Page

The **Infrastructure** page tracks well-known Kubernetes tooling components running in your cluster — things like Envoy Gateway, Prometheus, cert-manager, and more.

Infrastructure				
			Updated: 26/05/2026, 06:41:36	Send Report
Application	Namespace	Current Version	Latest Version	Status
Envoy Gateway	envoy-gateway-system	v1.8.0	v1.9.0-rc.0	Update Available
Kube State Metrics	ops-prometheus	v2.18.0	v2.19.0	Update Available
Prometheus Operator	ops-prometheus	v0.90.1	v0.91.0	Update Available
Reloader	reloader	v1.1.0	v1.4.17	Update Available
Sealed Secrets	sealed-secrets	0.32.2	v0.37.0	Update Available

Figure 3 — Infrastructure page with version comparison results

Reading the version table

COLUMN	DESCRIPTION
Application	The name of the component. Click the column header to sort alphabetically.
Namespace	The Kubernetes namespace where the deployment lives. Click the header to filter by namespace.
Current Version	The version currently running in your cluster, extracted from the deployment's container image tag.
Latest Version	The most recent version available from the project's GitHub releases or container registry.
Status	A badge summarising the comparison result. Click the column header to filter by status.

Status badges explained

BADGE	MEANING
✓ Up to Date ↗	The running version matches the latest available release. Click to open the release page.
⚠ Update Available ↗	A newer version has been published. Click to open the release page and see the changelog.
✗ Error	Beacon could not fetch the latest version. Usually a temporary network issue — it retries next day.
? Unknown	Version could not be extracted from the running image. May use a digest instead of a tag.

Infrastructure

Showing 1 of 6

Updated: 26/05/2026, 06:41:36

🔒 Send Report

Application ↗	Namespace ▾	Current Version	Latest Version	Status (1/4)
Metrics Server	kube-system	v1.8.23-14	N/A	? Unknown ↗

Infrastructure

Showing 5 of 6

Updated: 26/05/2026, 06:50:36

🔒 Send Report

Application ↗	Namespace ▾	Current Version	Latest Version	Status (1/4)
Envoy Gateway	envoy-gateway-system	v1.8.0	v1.9.0-rc.0	⚠ Update Available ↗
Kube State Metrics	ops-prometheus	v2.18.0	v2.19.0	⚠ Update Available ↗
Prometheus Operator	ops-prometheus	v0.90.1	v0.91.0	⚠ Update Available ↗
Reloader	reloader	v1.1.0	v1.4.17	⚠ Update Available ↗
Sealed Secrets	sealed-secrets	0.32.2	v0.37.0	⚠ Update Available ↗

Headlamp Plugins

Showing 1 of 3

Updated: 26/05/2026, 06:50:36

🔒 Send Report

Application ↗	Namespace ▾	Current Version	Latest Version	Status (1/4)
Flux Plugin	ops-headlamp	v0.6.0	v0.6.0	✓ Up to Date ↗

Figure 4 — The four status badge types

Sorting and filtering

The table supports interactive sorting and filtering to help you focus on what matters:

- **Sort by Application** — click the Application column header. An arrow indicates sort direction.
- **Filter by Namespace** — click the Namespace column header to open a checklist; tick the namespaces you want to see.

- **Filter by Status** — click the Status column header to show only specific badge types (e.g. "Update Available" only).

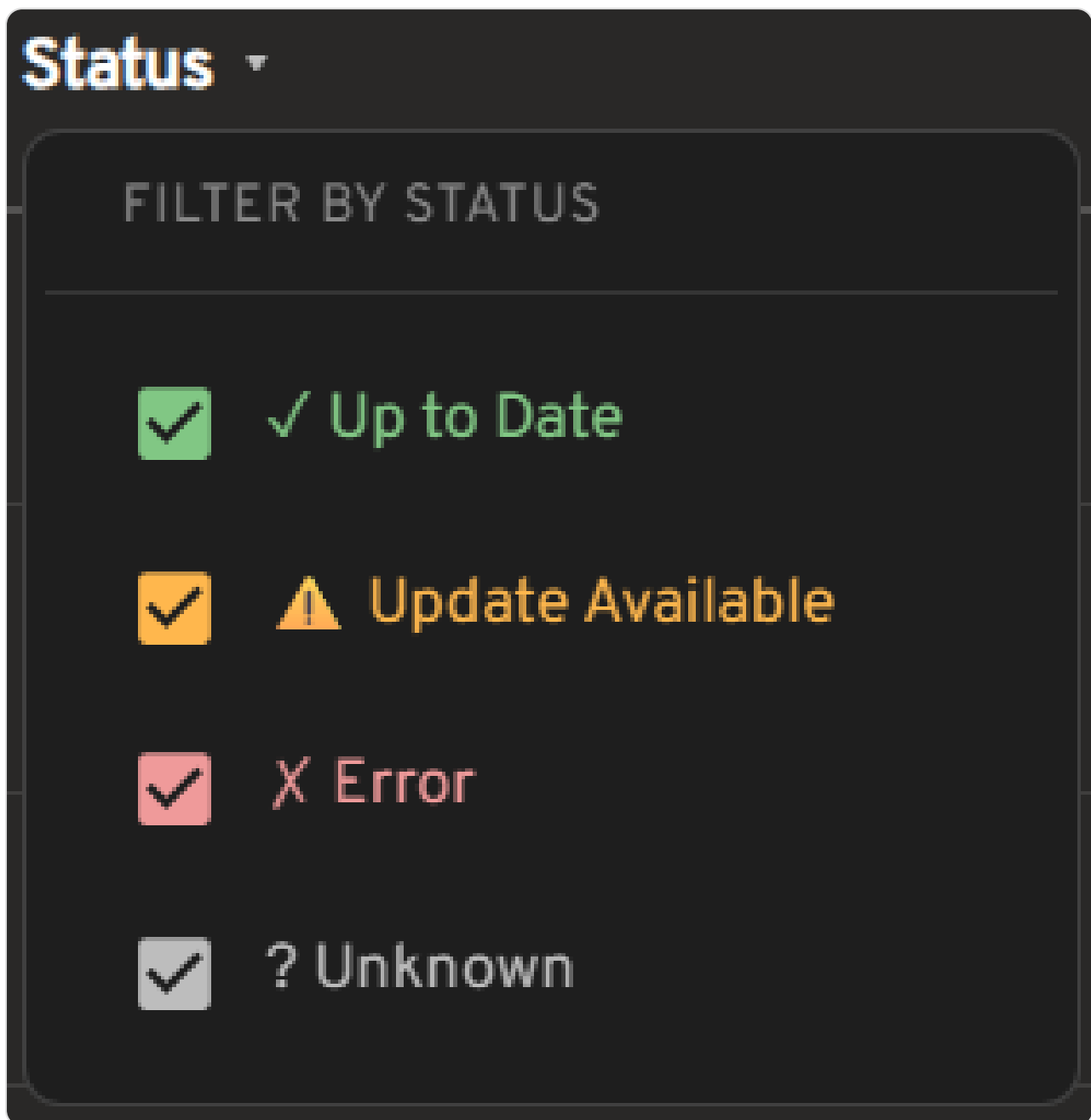


Figure 5 — Filtering by status to focus on components that need updates



When does Beacon update versions?

By default, Beacon fetches the latest versions once per day at 09:00. You can change the schedule in

Settings → Schedule

. The "Last updated" timestamp in the top-right shows when the most recent fetch ran.

4 Headlamp Plugins Page

The **Headlamp Plugins** page monitors Headlamp itself and the plugins installed alongside it as init containers — for example Flux, Kubescape, and Trivy.

Headlamp Plugins

Updated: 25/05/2026, 15:22:21

Send Report

Application ↑	Namespace ▾	Current Version	Latest Version	Status ▾
Flux Plugin	ops-headlamp	v0.6.0	v0.6.0	✓ Up to Date ↗
Headlamp	ops-headlamp	v0.41.0	v0.42.0	⚠ Update Available ↗
Kubescape Plugin	ops-headlamp	v0.10.8	v0.11.2	⚠ Update Available ↗

Figure 6 — Headlamp Plugins page

What is monitored

Beacon reads the Headlamp Kubernetes deployment to find all init containers. Each init container that matches a known plugin pattern is automatically detected. Headlamp itself is always included as an entry since it hosts the plugins.

The columns and badges are identical to the Infrastructure page. The **Namespace** column always shows `ops-headlamp` since all Headlamp init containers live in the same namespace as the Headlamp deployment.



Plugin not showing up?

If a plugin is installed but not appearing, go to

Settings → Headlamp Plugins

, click

Save Configuration

again to refresh the scope. Beacon re-reads the deployment's init containers on every save.

5 Applications Page



Pro Feature

The Applications page is only available in Beacon Pro and Enterprise editions. In the Free edition, this page displays a locked card explaining the feature.

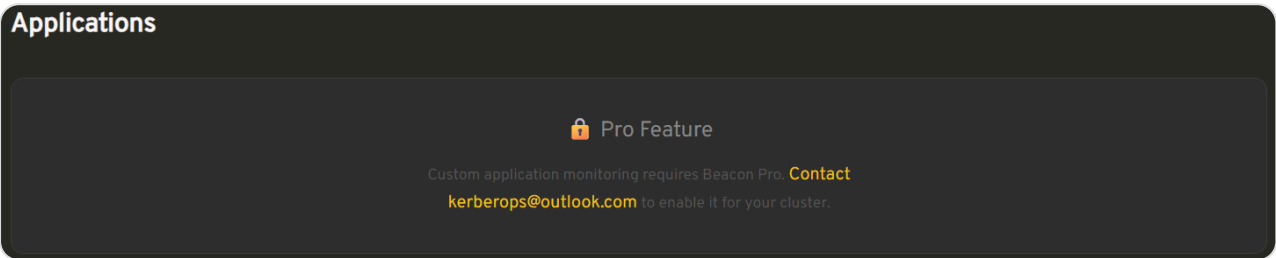


Figure 7 — Applications page in Free edition

In Beacon Pro, the Applications page monitors the custom workloads running in your cluster — your own services and internal APIs — with the same version comparison and status badge system used by the Infrastructure and Headlamp Plugins pages.

Contact your Beacon administrator to enquire about upgrading to Beacon Pro.

6 Settings

The **Settings** page is where you configure what Beacon monitors and when it checks for updates. Open it from **Beacon** → **Settings** in the sidebar.

The Settings page has five tabs. In the Free edition, two tabs are locked:

TAB	FREE	PRO
Applications	 locked	✓
Email & Reports	 locked	✓
Headlamp Plugins	✓	✓
Infrastructure	✓	✓
Schedule	✓	✓

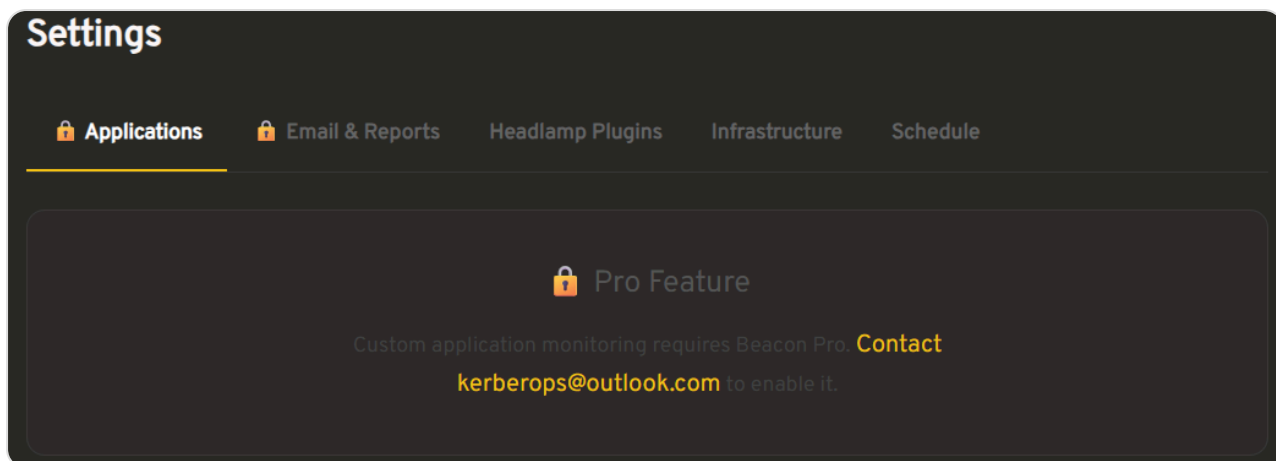


Figure 8 — Settings page tab bar in Free edition

Infrastructure tab

Use this tab to choose which infrastructure components Beacon monitors.

1 Open Settings → Infrastructure

Beacon automatically scans the cluster when this tab loads. Wait for the scan to complete (usually a few seconds).

2 Review DETECTED components

The green DETECTED section lists components found in your cluster. Each one has a toggle switch — turn it on to monitor it, off to exclude it.

3 Note the NOT DETECTED list

Components in the catalogue that are not installed in your cluster appear in a compact grey list at the bottom. These are automatically excluded.

4 Click Save Configuration

Beacon writes your selection and immediately triggers a version fetch. The Infrastructure page will reflect your changes within seconds.

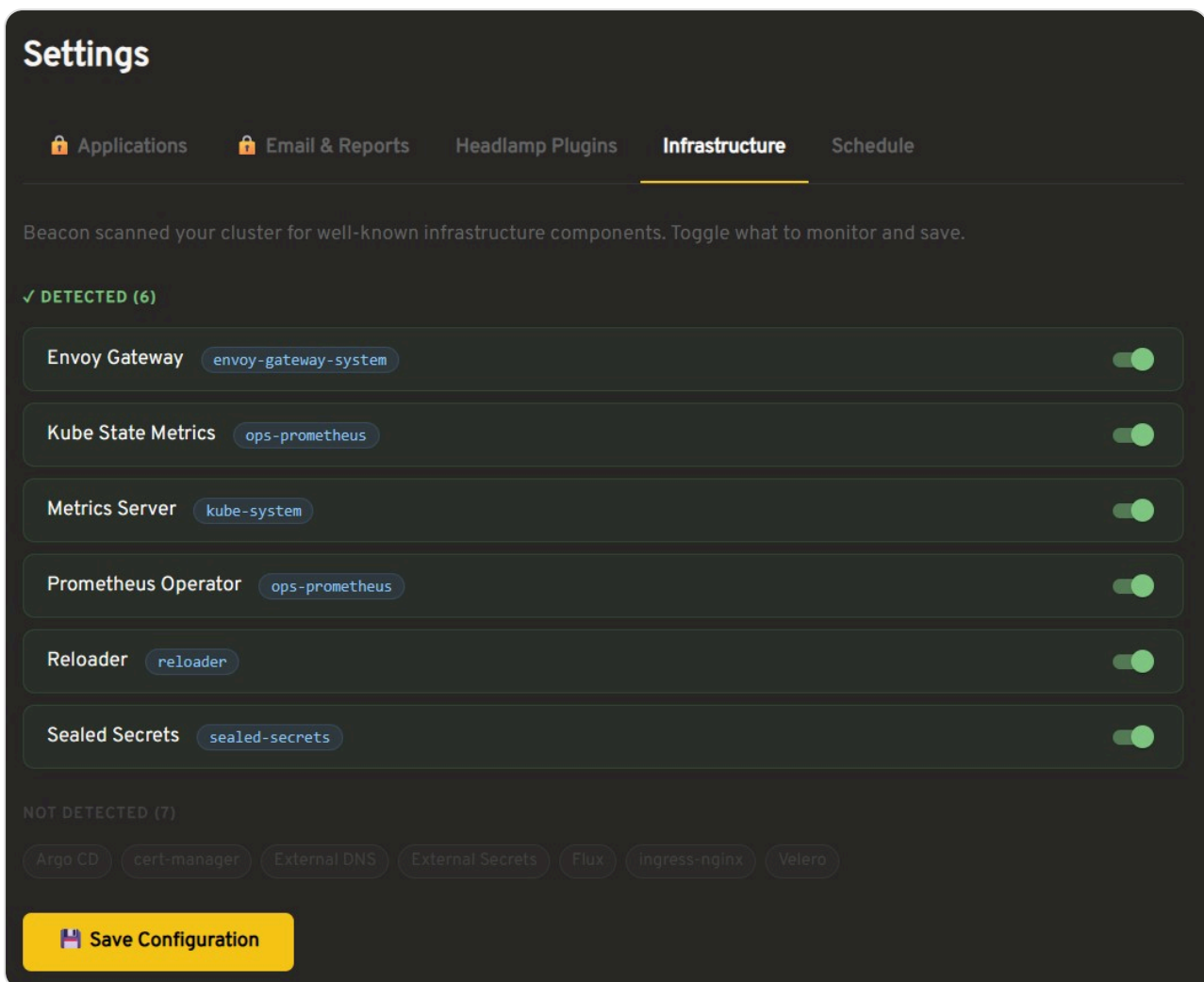


Figure 9 — Infrastructure tab with detected components and toggle switches

Headlamp Plugins tab

Use this tab to choose which Headlamp plugins Beacon monitors.

1 Open Settings → Headlamp Plugins

Beacon reads the Headlamp deployment's init containers. This is fast since it queries a single deployment.

2 Review DETECTED plugins

All detected plugins are listed with toggle switches. Headlamp itself always appears at the top.

3 Click Save Configuration

Beacon saves your selection and triggers a version fetch.

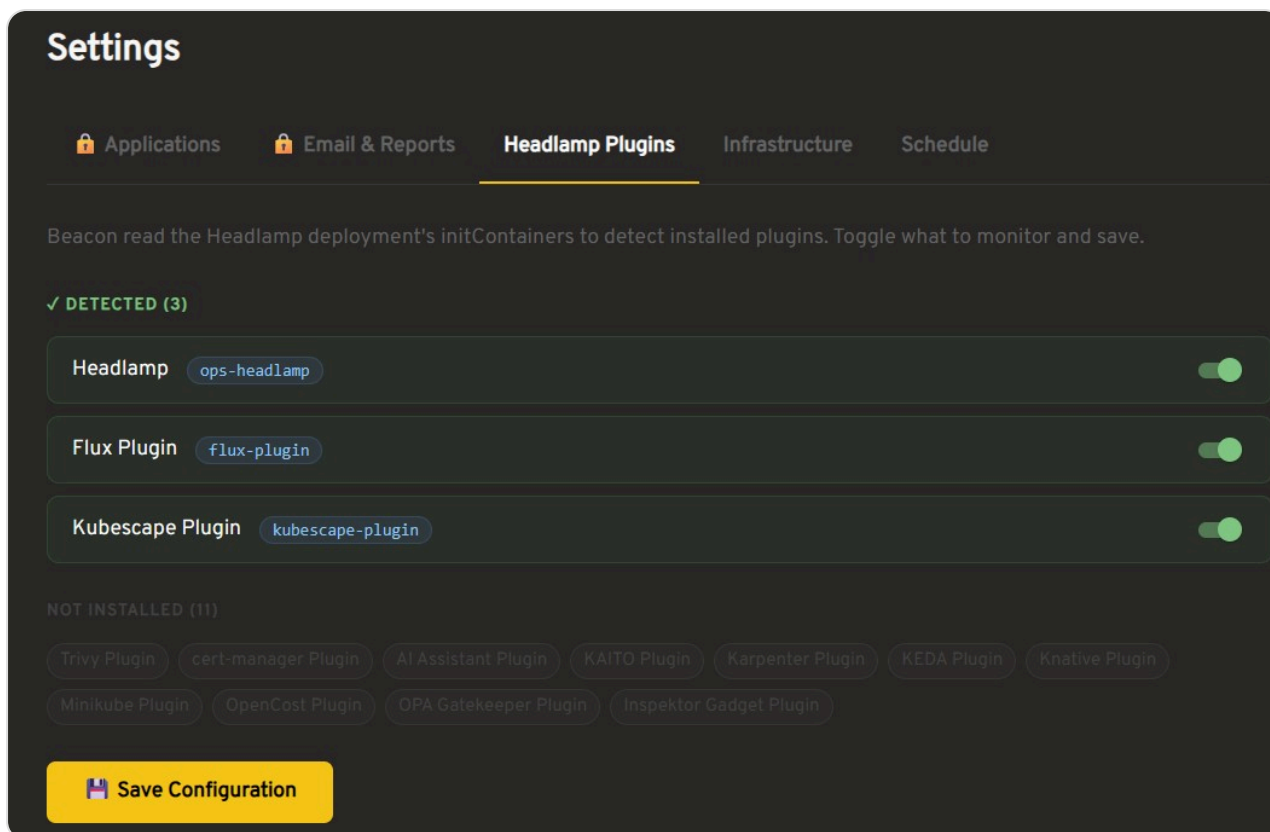


Figure 10 — Headlamp Plugins tab with detected plugins

Schedule tab

Use this tab to control *when* Beacon fetches the latest versions.

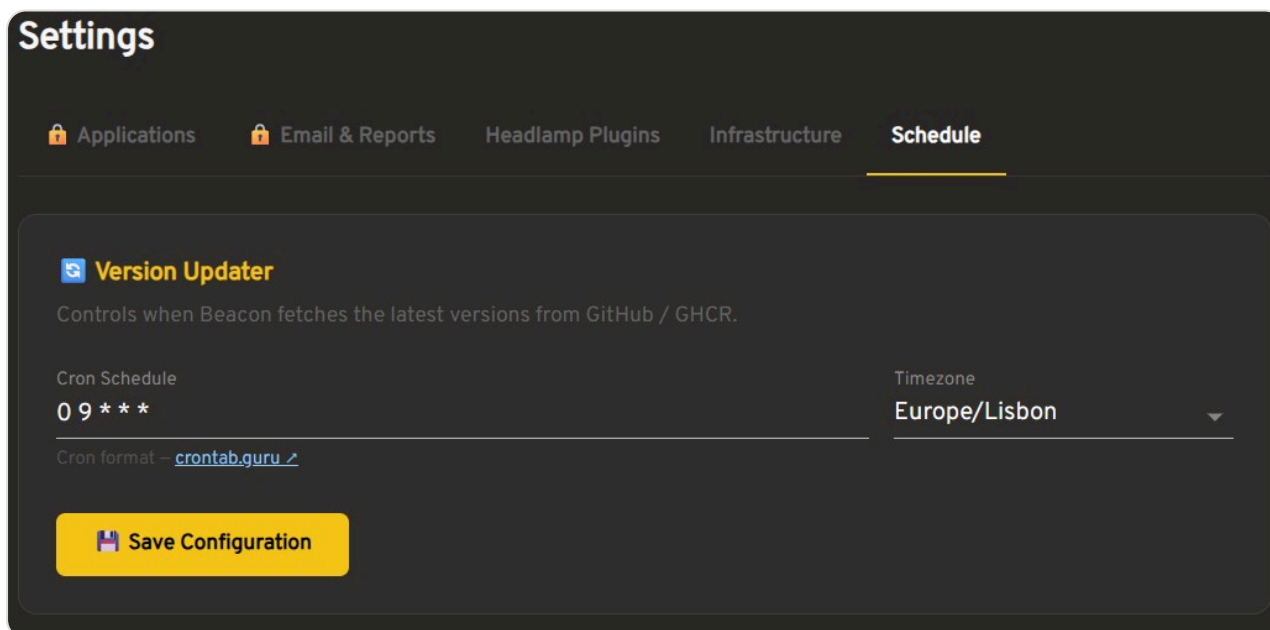


Figure 11 — Schedule tab — Version Updater section

VERSION UPDATER SECTION

FIELD	DESCRIPTION	DEFAULT
Cron Schedule	When the updater runs, in standard cron format: minute hour day-of-month month day-of-week Use the crontab.guru link to build your expression.	0 9 * * * (daily at 09:00)
Timezone	The timezone used to interpret the cron schedule. Select from the dropdown.	Europe/Lisbon

After changing either field, click  **Save Configuration**. The Kubernetes CronJob is updated immediately — the next run will use the new schedule.



Common schedule examples

— every day at 09:00

— every 6 hours

— every Monday at 09:00

— weekdays at 08:30



Daily Reports section (Pro only)

Below the Version Updater, Pro users see a Daily Reports section where they can enable/disable the automatic email report, set the schedule, timezone, and recipient addresses. This section is locked in Free edition.

7 Frequently Asked Questions

Versions show "Unknown" — what does that mean?

Beacon could not extract a version number from the running container's image tag. This usually happens when an image uses a digest (`sha256:...`) instead of a version tag, or when the image tag doesn't follow a standard version format. No action is required — the component is still monitored, and the badge will update when a parseable tag is deployed.

Versions haven't updated since yesterday

The updater runs on the schedule set in **Settings** → **Schedule**. If the "Last updated" timestamp in the top-right is older than expected, the CronJob may have been temporarily suspended. Contact your Kubernetes administrator to verify the `beacon-updater` CronJob is running in the `ops-headlamp` namespace.

A component I know is installed is not showing up

Beacon monitors only the components you have explicitly selected in **Settings**. If a component was installed after the last Settings save, go to Settings → Infrastructure (or Headlamp Plugins), verify the component appears in the DETECTED list, enable its toggle, and click Save.

The "Send Report" button is greyed out

The Send Report feature requires Beacon Pro. In the Free edition, the button is always displayed but locked. Contact your administrator to upgrade.

How do I add a new infrastructure component to Beacon's catalogue?

The catalogue of supported components is defined at the Beacon provider level. If you need a component added, contact your Beacon administrator. For standard open-source tools that are not yet in the catalogue, the Beacon team accepts requests.

Can Beacon see my application data?

No. Beacon only reads deployment metadata (image tags) and makes outbound requests to public APIs (GitHub, GHCR) to fetch version numbers. It does not access application logs, secrets, database contents, or any other workload data.